Response to Reviewer cAWZ:

Weakness 1:

Thank you very much for your valuable feedback. (1) The comparative experimental results with KD are shown in the Table B in **Official Comment**.

(2) We apologize for any confusion regarding the channel heatmap in Figure 2. The channel heatmap displayed is the predicted channel heatmap generated by the teacher model, DeepLabV3Plus-ResNet101. We have demonstrated visual results of our method compared to various state-of-the-art distillation methods in Figures 5-7.

Weakness 2:

Our method differs from the three reference papers provided by the reviewers as follows: First reference paper: They propose multi-scale distillation using feature pyramid layers, while our HRDKD performs multiple aggregations with RDKD Loss computed at each step.

Second reference paper: Their method involves self-decoupled ensemble distillation with focus on channel-wise feature maps, whereas our approach focuses on region-level features within divided blocks.

Third reference paper: They use a discriminator to separate target and non-target features and decouple non-target logits for distillation. In contrast, our method implicitly decouples the entire channel into target and non-target regions, aligning predictions on a block-level dimension using softmax calculations.

Weakness 3:

The comparative experimental results on the MS COCO and ADE 20k are shown in the Table A in **Official Comment**.

Weakness 4:

We rephrase the statement as follows: We aim to achieve state-of-the-art (SOTA) performance in semantic segmentation knowledge distillation methods, rather than in the field of semantic segmentation. We apologize for the typographical error in the description of Table 1. The correct entry should be ”S2:PSPNET-R18” instead of ”S2:PSPNET-R101”.

Weakness 5:

We have demonstrated the superior performance of our chosen loss weights in the domain of semantic segmentation knowledge distillation across the Cityscapes, Pascal VOC, Synapses, FLARE22, MS COCO and ADE20k datasets.

Weakness 6:

The reason why our proposed HRDKD performs worse than RDKD on the Pascal VOC dataset may be due to the presence of a large number of background pixels in the Pascal VOC labels, which leads to unfavorable effects during aggregation. However, for the fully pixel-level segmentation datasets, Cityscapes, Synapses, FLARE22, MS COCO and ADE20k datasets, HRDKD generally outperforms RDKD.

Weakness 7:

The reason we used the FCN architecture in medical image segmentation is its simplicity, and it is also easy to combine FCN with different lightweight backbones. This allows us to compare different architecture combinations. We have supplemented the experimental results using U-Net as a baseline, and the comparison experiments are shown in the Table C in **Official Comment**.

Question 1:

We have already explained this in Weakness 1.

Question 2:

The comparative experimental results with KD are shown in the Table B in **Official Comment**.

Question 3:

We have already explained this in Weakness 2.

Question 4:

We apologize for the inconvenience, but since the code for the mentioned paper is not publicly available, we are currently unable to directly compare our method with theirs. The comparative experimental results with TaT are shown in the table below.

Question 5:

The comparative experimental results on the MS COCO and ADE 20k are shown in the table below.

Question 6:

In Weakness 4, we explained the issues related to parameter settings and description errors, and we have made corrections and supplements to the tables and formulas. We apologize for any confusion caused during the review process.

Question 7:

We have already explained this in Weakness 5.

Question 8:

We have already explained this in Weakness 6.

Question 9:

We have already explained this in Weakness 7.